INTERNSHIP CANDIDATE RANKING SYSTEM

Algorithm, Flowchart, Pseudocode and Sample Output

1. Algorithm

Start the program.

Display the heading: Internship Candidate Ranking System.

Store the candidate names, marks, assessment scores and attendance percentages in lists.

Create an empty list named qualified.

For each candidate, check the eligibility conditions:

• Marks must be at least 60, assessment score must be at least 60, and attendance must be at least 75%.

• If all conditions are satisfied, calculate the total score using: total = marks × 0.4 + assessment × 0.4 + attendance × 0.2.

• Add the candidate's name, marks, assessment, attendance and total score to the qualified list.

• Otherwise, display that the candidate is not eligible.

Sort qualified candidates using these tie-breaking rules, in order: higher total score, higher assessment score, higher marks, and alphabetical order.

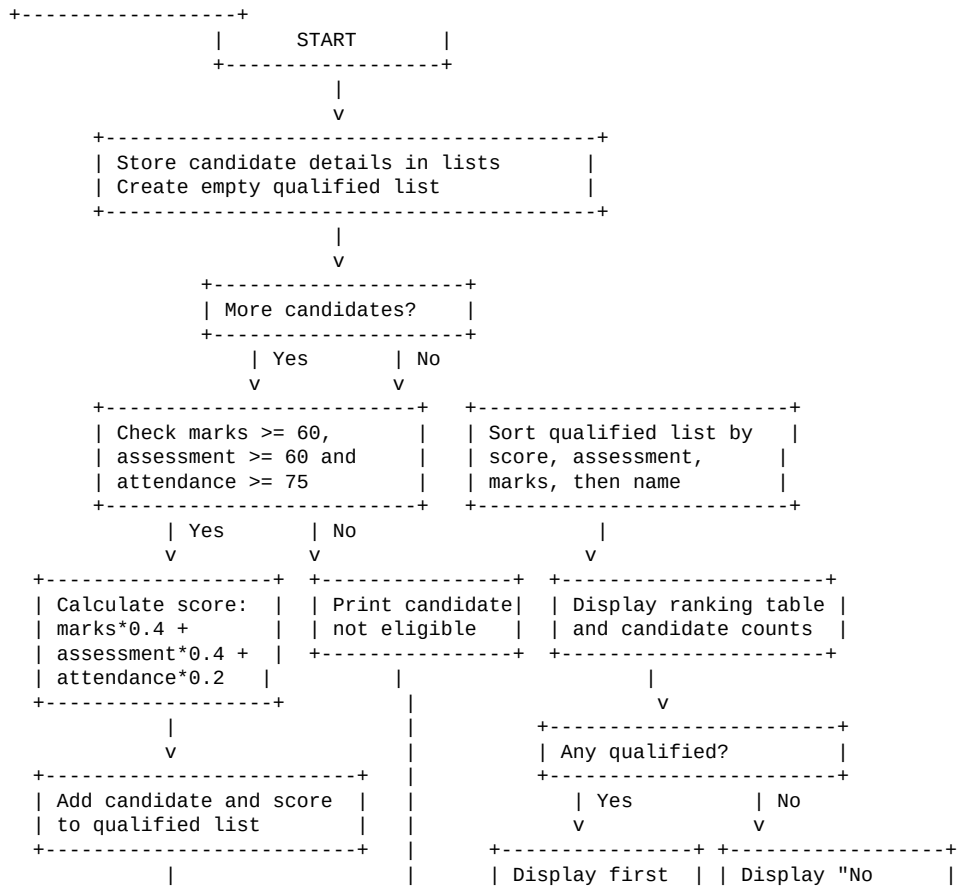
Display the ranking table of qualified candidates with their details and rounded total scores.

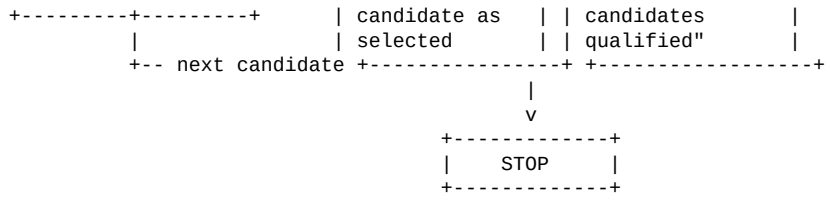
Display the total number of candidates, qualified candidates and candidates who are not qualified.

If at least one candidate qualified, display the first candidate in the sorted list as the selected candidate. Otherwise, display “No candidates qualified”.

Stop the program.

2. Flowchart





3. Pseudocode

```
BEGIN
    DISPLAY "INTERNSHIP CANDIDATE RANKING SYSTEM"

    STORE names, marks, assessment, attendance
    qualified = empty list

    FOR each candidate i
        IF marks[i] >= 60 AND assessment[i] >= 60
           AND attendance[i] >= 75 THEN

            total = marks[i] * 0.4
                    + assessment[i] * 0.4
                    + attendance[i] * 0.2

            ADD [name, marks, assessment, attendance, total]
                TO qualified
        ELSE
            DISPLAY names[i], "is not eligible"
        END IF
    END FOR

    SORT qualified BY:
        total score descending
        assessment score descending
        marks descending
        name alphabetical

    DISPLAY ranking of qualified candidates
    DISPLAY total, qualified and not qualified counts

    IF length(qualified) > 0 THEN
        DISPLAY qualified[0].name AS selected candidate
    ELSE
        DISPLAY "No candidates qualified"
    END IF
END
```

4. Sample Output

```
INTERNSHIP CANDIDATE RANKING SYSTEM
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--------- RANKING OF QUALIFIED CANDIDATES ---------
Rank  Name       Marks  Assessment  Attendance  Score
1     Sita       90     85          95          89.0
2     Priya      90     85          85          87.0
3     Ravi       85     80          90          84.0
4     Rahul      75     90          80          82.0
5     Arjun      80     70          75          75.0

Total Candidates: 5
Qualified Candidates: 5
Not Qualified: 0
Selected Candidate: Sita
```

Note: The sample output is based on the five candidate records in the supplied Python program. The total score uses weights of 40% for marks, 40% for assessment and 20% for attendance. All five candidates meet the eligibility conditions in this dataset.